Sports Med (2017) 47:193–195
DOI 10.1007/s40279-016-0580-3

LEADING ARTICLE

Determining Strength: A Case for Multiple Methods of Measurement

Samuel L. Buckner[1] · Matthew B. Jessee[1] · Kevin T. Mattocks[1] · J. Grant Mouser[1] · Brittany R. Counts[1] · Scott J. Dankel[1] · Jeremy P. Loenneke[1]

Published online: 5 July 2016
© Springer International Publishing Switzerland 2016

Abstract Muscle strength is often measured through the performance of a one-repetition maximum (1RM). However, we that feel a true measurement of 'strength' remains elusive. For example, low-load alternatives to traditional resistance training result in muscle hypertrophic changes similar to those resulting from traditional high-load resistance training, with less robust changes observed with maximal strength measured by the 1RM. However, when strength is measured using a test to which both groups are 'naive', differences in strength become less apparent. We suggest that the 1RM is a specific skill, which will improve most when training incorporates its practice or when a lift is completed at a near-maximal load. Thus, if we only recognize increases in the 1RM as indicative of strength, we will overlook many effective and diverse alternatives to traditional high-load resistance training. We wish to suggest that multiple measurements of strength assessment be utilized in order to capture a more complete picture of the adaptation to resistance training.

✉ Jeremy P. Loenneke
jploenne@olemiss.edu

[1] Kevser Ermin Applied Physiology Laboratory, Department of Health, Exercise Science, and Recreation Management, The University of Mississippi, 231 Turner Cente, University, MS 38677, USA

Key Points

- Strength is typically assessed through a one-repetition maximum (1RM) test.
- The 1RM is a specific skill, which will improve most when training closely reflects the 1RM test.
- Multiple measurements of strength should be utilized to best capture strength adaptation.

1 Introduction

Attempts to measure the force-producing capabilities of the musculature are often assessed through different performance measures. For example, isometric [1], isokinetic [2] and one-repetition maximum (1RM) tests [1–3] are all used to assess strength adaptation. However, we feel that a true measurement of 'strength' remains elusive. For example, low-load resistance training has been shown to result in muscle hypertrophic changes similar to those resulting from traditional high-load resistance training, with less robust changes observed with maximal strength measured by a 1RM [1, 3]. However, when strength is measured using a test to which both groups are 'naive', differences in strength become less apparent [1, 2, 4]. This concept is best illustrated by the work of Mitchell et al. [1], who showed that training with higher loads (80 % of 1RM) led to greater increases in 1RM strength than training with lower loads (30 % of 1RM). However, similar increases were observed in maximal isometric strength, maximal instantaneous power output and the rate of force development. Similarly, Martin-Hernandez et al. [2] found comparable

increases in isokinetic peak torque at 60 and 180 °/s between 5 weeks of traditional high-load resistance exercise and 5 weeks of low-load resistance exercise with blood flow restriction, with greater increases in 1RM strength observed in the high-load training group. This suggests that the group that trained near a 1RM performed better at the 1RM test, which may have been due to specificity and the fact that the low-load training group had never been exposed to lifting maximally through a range of motion. Interestingly, if a 1RM were the only strength test employed, the low-load training group would not appear to elicit a comparable performance benefit. Thus, a 1RM test may not capture all of the strength adaptations achieved in a resistance training programme, and the 'strength' benefit derived from training at submaximal loads may be overlooked if a 1RM test is the only strength test employed.

2 The Evidence

Although the past several years of research have provided a wealth of evidence regarding the efficacy of alternatives to traditional high-load resistance training [5], the recommendations of governing bodies still maintain that intensities near 60–80 % of 1RM are necessary for increases in strength [6, 7]. However, we believe that the evidence behind the current recommendations supports our assertion that strength is subjective in nature and is dependent on the test used to assess it. For example, the work of Campos et al. [8] found greater increases in 1RM strength in a group training with a high relative load (3–5RM) than in a group training with a lower relative load (20–28RM); however, the 20–28RM group performed better on a test of repetitions to failure at 60 % of 1RM. The authors cited these findings as evidence of the strength–endurance continuum—an assertion that may undermine the adaptations occurring with lower loads. It appears that the strength–endurance continuum may reflect the specificity of the test employed, which in most cases is a 1RM. These data appear to provide strong evidence that specificity may be the best predictor of performance on a strength test and may be more attributable to the training programme that more closely resembles the test.

The importance of specificity regarding resistance exercise has been known for quite some time. Stone et al. [9] presented a theoretical model of strength training in 1982, suggesting that specificity is the most important component in a resistance training program. On the basis of the recommendations, low-load resistance training would fall under 'local muscular endurance' despite its effectiveness in increasing muscle size and strength (depending on the test) being similar to that of high-load training [1, 2, 5]. Considering that the change in muscle size explains only a small portion of the variance in the change in strength [10], this suggests that these strength differences are likely related to repeated motor patterns. This is likely a result of skill acquisition and/or repeated practice of the movement [11]. This was elegantly demonstrated in the work of Morton et al. [12], which showed that multiple exposures to a 1RM during a low-load resistance training programme could largely abolish the difference in 1RM strength typically observed between high-load and low-load training modalities. Thus, periodic practice/exposure to the lift that is tested appears to be enough to negate the differences in strength that are observed.

3 Conclusion

As we continue to better understand adaptations to resistance training, it is essential to revisit—and in some cases challenge—long-accepted dogma. Regarding strength, there seems to be resistance to acknowledging the efficacy of low-load alternatives to traditional resistance training and their ability to augment strength similarly to high-load training. 1RM is a specific skill that will improve most when training more closely reflects the test. Notably, 1RM testing may have greater relevance in some populations (e.g. powerlifters) if an improvement in 1RM skill is the sole purpose of that intervention. However, if we continue to recognize only increases in 1RM as indicative of strength, we will overlook many effective and diverse alternatives to traditional high-load resistance training. When other physiological variables, such as muscle damage, are observed, multiple indirect measurements are typically taken, such as measures of soreness, decrements in torque production and swelling [13]. A similar approach may be warranted with strength assessment. This contention is further supported by the findings of Churchward-Venne et al. [14], who found that there were no 'non-responders' to resistance training when multiple measurements (leg press, leg extension, and chair-raise time) were examined, since all participants improved in some way. Focusing on only one measurement would have resulted in some individuals being labelled as 'non-responders'. Future research should further examine strength measurement and its ability to reflect adaptation to both high-load and low-load resistance training.

Compliance with Ethical Standards

Funding No external sources of funding were used in the preparation of this article.

Conflict of interest Samuel Buckner, Matthew Jessee, Kevin Mattocks, Grant Mouser, Brittany Counts, Scott Dankel and Jeremy Loenneke declare that they have no conflicts of interest that are relevant to the content of this article.

References

1. Mitchell CJ, Churchward-Venne TA, West DW, et al. Resistance exercise load does not determine training-mediated hypertrophic gains in young men. J Appl Physiol. 2012;113(1):71–7.
2. Martin-Hernandez J, Marin PJ, Menendez H, et al. Muscular adaptations after two different volumes of blood flow-restricted training. Scand J Med Sci Sports. 2013;23(2):e114–20.
3. Ogasawara R, Loenneke JP, Thiebaud RS, et al. Low-load bench press training to fatigue results in muscle hypertrophy similar to high-load bench press training. Int J Clin Med. 2013;4(02):114.
4. Ward J, Fisk GH. The difference in response of the quadriceps and the biceps brachii muscles to isometric and isotonic exercise. Arch Phys Med Rehabil. 1964;45:614–20.
5. Ozaki H, Loenneke JP, Buckner SL, et al. Muscle growth across a variety of exercise modalities and intensities: contributions of mechanical and metabolic stimuli. Med Hypotheses. 2016;88:22–6.
6. Garber CE, Blissmer B, Deschenes MR, et al. American College of Sports Medicine position stand. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: guidance for prescribing exercise. Med Sci Sports Exerc. 2011;43(7):1334–59.
7. Essentials of strength training and conditioning: National Strength and Conditioning Association. Baechle TR, Earle RW, editors. Champaign: Human Kinetics, 3rd edition; 2008. p. 401.
8. Campos GE, Luecke TJ, Wendeln HK, et al. Muscular adaptations in response to three different resistance-training regimens: specificity of repetition maximum training zones. Eur J Appl Physiol. 2002;88(1–2):50–60.
9. Stone MH, O'Bryant H, Garhammer J, et al. A theoretical model of strength training. Strength Cond J. 1982;4(4):36–9.
10. Ahtiainen JP, Walker S, Peltonen H, et al. Heterogeneity in resistance training-induced muscle strength and mass responses in men and women of different ages. Age (Dordr). 2016;38(1):1–13.
11. Bernardi M, Solomonow M, Nguyen G, et al. Motor unit recruitment strategy changes with skill acquisition. Eur J Appl Physiol Occup Physiol. 1996;74(1–2):52–9.
12. Morton RW, Oikawa SY, Wavell CG, et al. Neither load nor systemic hormones determine resistance training-mediated hypertrophy or strength gains in resistance-trained young men. J Appl Physiol. 2016. doi:10.1152/japplphysiol.00154.2016 (Epub 12 May 2016).
13. Loenneke JP, Thiebaud RS, Abe T. Does blood flow restriction result in skeletal muscle damage? A critical review of available evidence. Scand J Med Sci Sports. 2014;24(6):e415–22.
14. Churchward-Venne TA, Tieland M, Verdijk LB, et al. There are no nonresponders to resistance-type exercise training in older men and women. J Am Med Dir Assoc. 2015;16(5):400–11.